Roberto Bruno



Postdoctoral Researcher | Information Theory, Algorithms, Data Compression

Location: Bagnoli Irpino (AV), Italy | Phone: +39 345 856 7214
Email: roberto250398@gmail.com | rbruno@unisa.it
LinkedIn: linkedin.com/in/robertobruno11001 | GitHub: github.com/R11001
ORCID: 0000-0001-6039-2075

PROFESSIONAL SUMMARY

Computer Scientist with a Ph.D. and postdoctoral experience in designing efficient algorithms for data compression, search and optimization problems. Author of 12 peer-reviewed publications in top venues, including IEEE Transactions on Information Theory. Strong background in algorithm design and analysis, computational complexity, approximation algorithms and mathematical modeling. Hands-on programming experience in C/C++, Python and Java, and in parallel and high-performance computing (CUDA, SYCL, OpenMP, MPI), with experience in Git and Docker. Proven ability to solve complex problems, communicate technical results clearly and work in international teams, including a research stay at Okinawa Institute of Science and Technology (OIST), Japan.

WORK EXPERIENCE

Postdoctoral Researcher
University of Salerno, Department of Computer Science, Fisciano (SA), Italy | Jul 2026 – Present
Research on differential privacy, information theory and computational complexity.

Ph.D. Researcher in Computer Science
University of Salerno, Fisciano (SA), Italy | Nov 2022 – Mar 2026
Supervisor: Prof. Ugo Vaccaro
Research on the duality between searching and coding, focusing on alphabetic codes, prefix codes and binary search trees. Established novel entropic bounds on the average length of binary codes, and thus on the complexity of the corresponding search algorithms, and designed linear-time algorithms for their construction. Proved NP-hardness results and developed near-optimal approximation algorithms for information bottleneck, dimension reduction and data compression problems. Results published in IEEE Transactions on Information Theory, Information Processing Letters, Entropy, Algorithms and AIMS Mathematics, and presented at IEEE ISIT.

Visiting Research Student
Okinawa Institute of Science and Technology (OIST), Okinawa, Japan | May 2025 – Nov 2025
Supervisor: Prof. Amedeo Roberto Esposito
Applied information measures, in particular Rényi divergence, to the hypothesis testing problem to derive novel finite-sample bounds. Results accepted for publication at IEEE ISIT 2026.

Teaching Assistant – Algorithm Design
University of Salerno, Fisciano (SA), Italy | Mar 2024 – Sep 2024
Teaching assistant for a bachelor's course in algorithm design.

EDUCATION

Ph.D. in Computer Science
University of Salerno, Fisciano (SA), Italy | 2022 – 2026

- Thesis (Information Theory): From Search to Coding (and Back): Two Sides of the Same Coin. Supervisor: Prof. Ugo Vaccaro.

Master's Degree in Computer Science (Laurea Magistrale) – 110/110 cum laude
University of Salerno, Fisciano (SA), Italy | 2020 – 2022

- Thesis: Application of Information Theory functionals to dimension reduction of probability distributions.
- Relevant coursework: Information Theory, Data Compression, Optimization Methods (Linear and Integer Programming), Advanced Algorithms (exact, approximation, randomized, online and distributed algorithms), High-Performance Computing, Distributed Architectures (P2P protocols, leader election, consensus).

Bachelor's Degree in Computer Science (Laurea Triennale) – 110/110 cum laude
University of Salerno, Fisciano (SA), Italy | 2017 – 2020

- Thesis: Exact Exponential Algorithms.

SKILLS

- **Programming Languages:** C, C++, Python, Java, JavaScript, MATLAB, SQL.
- **Parallel and High-Performance Computing:** CUDA, SYCL, OpenMP, MPI, GPU programming, performance portability.

- **Tools and Technologies:** Git, Docker, relational databases, NoSQL databases (MongoDB), LaTeX.
- **Technical Expertise:** Information Theory, Data Compression, Coding Theory, Algorithm Design and Analysis, Data Structures, Approximation Algorithms, Computational Complexity, Optimization, Dimension Reduction, Hypothesis Testing, Distributed Systems.
- **Soft Skills:** Scientific writing, teaching, research project management, time management, teamwork, international collaboration.
- **Research Interests:** Information Theory, Algorithms and Data Structures, Game Theory, Artificial Intelligence.

LANGUAGES

- **Italian:** Native.
- **English:** C1 (reading, writing), B2 (listening, speaking). Certificate: Centro Linguistico di Ateneo, University of Salerno, June 2021.

PROJECTS

- **SYCL Performance Portability** (github.com/R11001/SYCL_Performance_Portability): performance portability study of matrix multiplication implementations on AMD CPU and NVIDIA GPU using SYCL (High-Performance Computing course).
- **Auction Mechanism** (github.com/R11001/ADC_Auction-Mechanism): design and development in Java of an auction mechanism on a peer-to-peer (P2P) network using the TomP2P library.
- **Toy Compiler**: source-to-source compiler covering all stages from lexical analysis to intermediate code generation (team project).
- **MyTutor**: software engineering project to digitalize the request procedure for teaching assistance activities at the Department of Computer Science, University of Salerno.

PUBLICATIONS

Journal Articles

- R. Bruno, R. De Prisco, U. Vaccaro. Optimal Average-Case Binary Search with Outcome-Dependent Costs. Information Processing Letters, 2026. DOI: 10.1016/j.ipl.2026.106640
- R. Bruno, U. Vaccaro. NP-Hardness and Approximation Algorithms for Constrained Entropy Maximization. IEEE Transactions on Information Theory, 2025. DOI: 10.1109/TIT.2025.3644150
- R. Bruno, R. De Prisco, A. De Santis, U. Vaccaro. Bounds and Algorithms for Alphabetic Codes and Binary Search Trees. IEEE Transactions on Information Theory, 2024. DOI: 10.1109/TIT.2024.3428699
- R. Bruno. Hardness and Approximability of Dimension Reduction on the Probability Simplex. Algorithms (MDPI), 2024. DOI: 10.3390/a17070296. Editor's Choice.
- R. Bruno, U. Vaccaro. Entropic Bounds on the Average Length of Codes with a Space. Entropy (MDPI), 2024. DOI: 10.3390/e26040283
- R. Bruno, U. Vaccaro. A Note on Equivalent Conditions for Majorization. AIMS Mathematics, 2024. DOI: 10.3934/math.2024419
- R. Bruno, U. Vaccaro. An Information Theoretic Treatment of Yager's Probability Distribution Negation. Soft Computing, 2026. DOI:10.1007/s00500-026-11413-9

Conference Papers

- R. Bruno, A. Vandenbroucque, A. R. Esposito. A Finite-Sample Strong Converse for Binary Hypothesis Testing via (Reverse) Rényi Divergence. IEEE International Symposium on Information Theory (ISIT 2026), Guangzhou, China.
- R. Bruno, U. Vaccaro. Constrained Maximum Entropy Contiguous Aggregations. IEEE ISIT 2026, Guangzhou, China.
- R. Bruno, A. A. Rescigno, U. Vaccaro. Robust Shift-Invariant Superimposed Codes. IEEE ISIT 2026, Guangzhou, China.
- R. Bruno, R. De Prisco, U. Vaccaro. Optimal Binary Variable-Length Codes with a Bounded Number of 1's per Codeword: Design, Analysis, and Applications. IEEE ISIT 2025, Ann Arbor, Michigan, USA.
- R. Bruno, R. De Prisco, U. Vaccaro. Old and New Results on Alphabetic Codes. Workshop on Information Theory and Related Fields, Lecture Notes in Computer Science (Springer). DOI: 10.1007/978-3-031-82014-4_7

I authorize the processing of my personal data for recruitment purposes in accordance with Regulation (EU) 2016/679 (GDPR).